

Definition

A $100(1 - \alpha)\%$ CI for the mean μ of a normal population when σ is known is given by:

$$\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

Example: A cargo company

A cargo company aims to have a mean delivery time of 5 hours for local deliveries.

They believe that delivery times are normal with $\sigma=2.4$.

They recorded 10 delivery times as:

7.3, 3.3, 4.7, 6.8, 10.1, 5.7, 6.4, 4.2, 3.1, 8.9

95%CI on the true mean delivery time?

$$\bar{x} = 6.05$$

$$95\%CI: (4.5625, 7.5375)$$

$$100(1 - \alpha)\% \text{ CI for } \mu \\ \left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

$$z_{0.025} = 1.96$$

The correct interpretation is by many samples via frequency...

Assume that M samples are collected, each having n observations in it.

We use the above formula to calculate a confidence interval for each of these samples.

Here is a description, and you will do this in Lab 4 yourself...

M samples, each with n observations,
and $1-\alpha$ as the confidence level

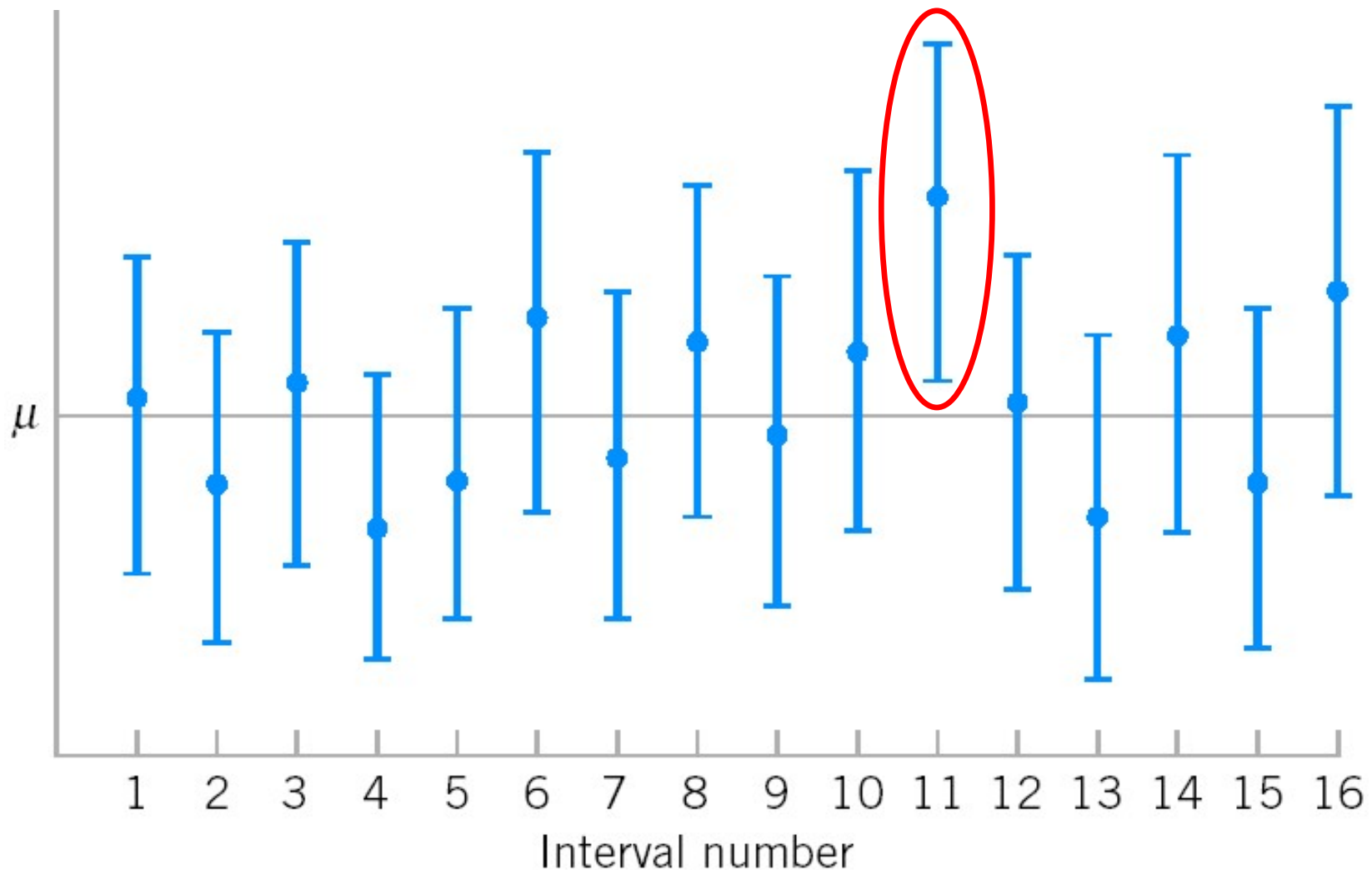
	Observations				Mean	lower bnd	upper bnd
Sample	1	2	...	n	$\bar{x}$	l	u
1	x_1^1	x_2^1	...	x_n^1	$\bar{x}^1$	l^1	u^1
2	x_1^2	x_2^2	...	x_n^2	$\bar{x}^2$	l^2	u^2
M-1	x_1^{M-1}	x_2^{M-1}	...	x_n^{M-1}	$\bar{x}^{M-1}$	l^{M-1}	u^{M-1}
M	x_1^M	x_2^M	...	x_n^M	$\bar{x}^M$	l^M	u^M

How many intervals will
cover the real μ ?

$$\bar{x}^M - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$\bar{x}^M + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Confidence Interval on the Mean of a Normal Distribution, Variance Known



Repeated construction of a confidence interval for μ .

Confidence Interval on the Mean of a Normal Distribution, Variance Known

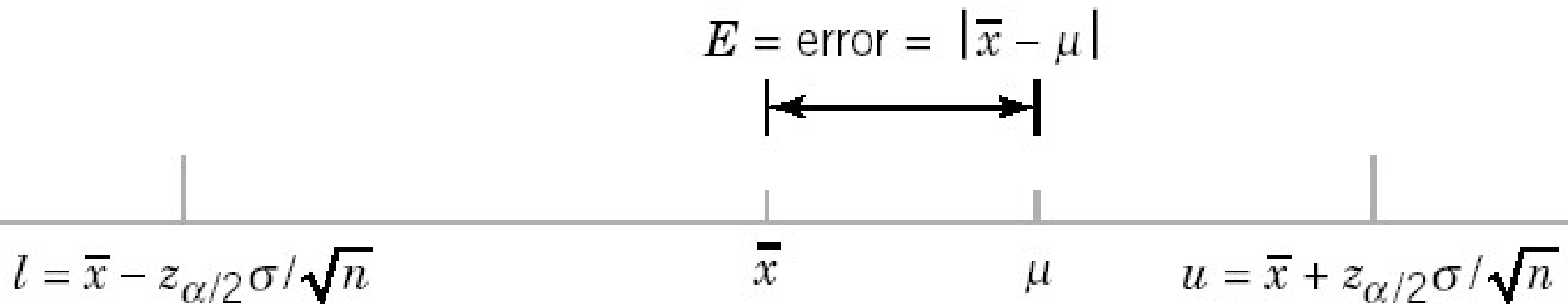
Interpreting a Confidence Interval

- The confidence interval is a **random interval**.
- The appropriate interpretation of a confidence interval (for example on μ) is: The observed interval $[l, u]$ brackets the true value of μ , with confidence $100(1-\alpha)$.

Confidence Interval on the Mean of a Normal Distribution, Variance Known

Confidence Level and Precision of Error

The length of a confidence interval is a measure of the **precision** of estimation.



Error in estimating μ with $\bar{x}$.

$$100(1 - \alpha)\% \text{CI: } \left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

Confidence Level & Precision

Confidence Level

Width of the Interval

$100(1 - \alpha)\%$

$$2z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

95%

$$2(1.96) \frac{\sigma}{\sqrt{n}}$$

99%

$$2(2.58) \frac{\sigma}{\sqrt{n}}$$

100%

∞

Confidence Level, Precision & Choice of Sample Size

For fixed sample size: If confidence level is increased, the width of CI is also increased, so that the precision or accuracy decreases if the reliability of CI is increased.

To have both high confidence & high precision, the only way is increasing the sample size.

Precision & Choice of Sample Size

What is the width of 100(1- α)% CI? $2z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

Let w be the desired width (so the precision you aim) of CI.

What is the required n ? $n \geq \left(2z_{\alpha/2} \frac{\sigma}{w}\right)^2$

Example: A cargo company

A cargo company aims to have a mean delivery time of 5 hours for local deliveries.

They believe that delivery times are normal with $\sigma=2.4$.

They recorded 10 delivery times as:

7.3, 3.3, 4.7, 6.8, 10.1, 5.7, 6.4, 4.2, 3.1, 8.9

If the desired width of 95%CI is 2, how many more delivery times should be recorded?

$$\bar{x} = 6.05$$

To summarize:

$X_1, \dots, X_n \sim N(\mu, \sigma)$: random sample

Assumptions: μ is unknown and σ is known.

$$100(1-\alpha)\% \text{CI} : \left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

- Interpretation?
- Precision, confidence, sample size?

The same setting:

$X_1, \dots, X_n \sim N(\mu, \sigma)$: random sample

Assumptions: μ is unknown and σ is known.

Can we define one-sided CIs?

Example: A cargo company

A cargo company aims to have a mean delivery time **of at most 5 hours** for local deliveries.

They believe that delivery times are normal with $\sigma=2.4$.

They recorded 10 delivery times as:

7.3, 3.3, 4.7, 6.8, 10.1, 5.7, 6.4, 4.2, 3.1, 8.9

Can we check whether the company achieves its aim at 95% confidence?

$$\bar{x} = 6.05$$

One-sided CIs

$P(L < \mu) = 1 - \alpha$, where L is a random variable.

means “probability that L is less than μ is $1-\alpha$ ”.

Now assume that an observation is made on the random variable L so that $L=l$.

$P(l < \mu) = 1 - \alpha$,

where l is an observation of the random variable L .

means “with $100(1-\alpha)\%$ confidence level μ is greater than l ”

... and α is called the significance level.

$$100(1-\alpha)\% \text{CI} : \left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

One-sided CIs

$X_1, \dots, X_n \sim N(\mu, \sigma)$: random sample

Assumptions: μ is unknown and σ is known.

Can you find a lower bound on the population mean μ with $1-\alpha$ confidence?

$$\bar{x} - z_{\alpha} \frac{\sigma}{\sqrt{n}}$$

Can you find an upper bound on the population mean μ with $1-\alpha$ confidence?

$$\bar{x} + z_{\alpha} \frac{\sigma}{\sqrt{n}}$$

Example: A cargo company

A cargo company aims to have a mean delivery time **of at most 5 hours** for local deliveries.

They believe that delivery times are normal with $\sigma=2.4$.

They recorded 10 delivery times as:

7.3, 3.3, 4.7, 6.8, 10.1, 5.7, 6.4, 4.2, 3.1, 8.9

Can we check whether the company achieves its aim at 97.5% confidence? How about 95%? At 90%?

$$\bar{x} = 6.05$$

Significance	Lower bound	Upper bound
90%	5.08	7.02
95%	4.80	7.30
97.5%	4.56	7.54

$$z_{0.05}=1.645$$

$$z_{0.10}=1.28$$

$$95\%CI: (4.5625, 7.5375)$$

Confidence Interval for the Mean of a r.v. with a large sample

$X_1, \dots, X_n$: random sample

Assumptions:

- n is very large. $\longrightarrow \bar{X} \overset{approx}{\sim} N(\mu, \sigma/\sqrt{n})$ by CLT.
- σ is known.

CI is the same as in normal r.v.s: $\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$

But σ is almost never known!!!

Confidence Interval for the Mean of a r.v. with Unknown Variance:

$X_1, \dots, X_n$: random sample (no specification on their distribution)

Assumptions:

- n is very large. $\longrightarrow \bar{X} \overset{approx}{\sim} N(\mu, \sigma/\sqrt{n})$ by CLT.
- both μ and σ are unknown.

σ will be estimated by sample standard deviation S .

The change in C. I.?

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \longrightarrow \text{more variable compared to } \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

but still OK, since n is large!!!

Proposition

If n is sufficiently large:

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \underset{\sim}{approx.} N(0,1)$$

so that

$$\left(\bar{x} - z_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{s}{\sqrt{n}} \right)$$

is a large-sample interval for μ with a confidence level approximately $100(1-\alpha)\%$. This formula is valid regardless of the population distribution.

Rule of thumb

$n > 40$ is generally sufficient for all kinds of random variables. To “fine-tune” this rule for specific probability distributions, you can run simulations.

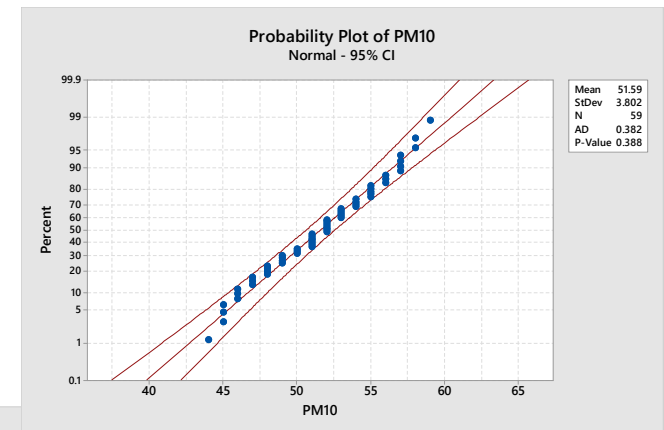
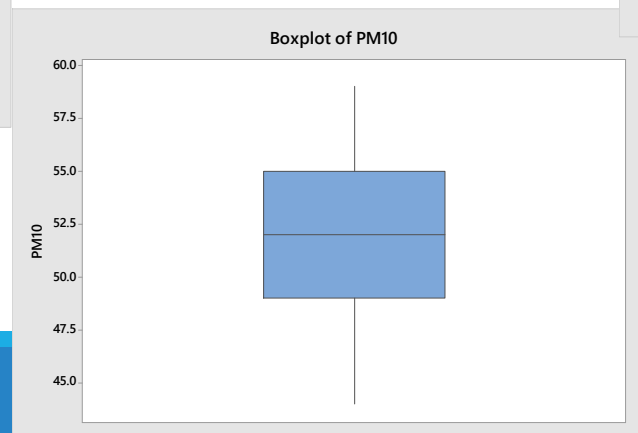
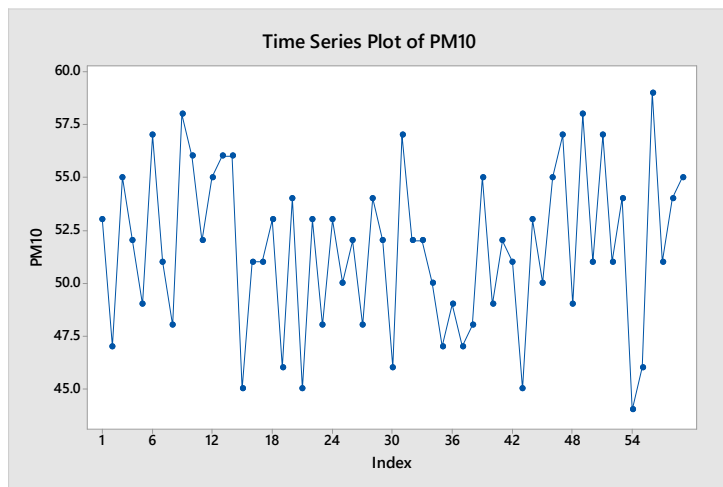
More conservative than CLT because of the additional variability introduced by using S instead of σ .

You can experiment with “ $n > 40$.”

Class exercise

In the small city, where green environment supporters claim that the air is heavily polluted, daily PM10 levels for the last two months (January and February 2018) have been recorded.

$$N=59, \bar{x} = 51.59, s=3.802$$



Class exercise

Green environment supporters claim that the average daily PM10 level exceeds the upper limit of 50mg. Evaluate this claim using appropriate confidence intervals or bounds using the data above.

Assumptions:

1. When the time series plot is checked, we see nothing but random fluctuations. Therefore, we can assume that the data is a random sample.
2. Both the stem-and-leaf diagram and the boxplot show that the data look symmetric. Moreover, the data is around the 45⁰ line in the probability plot, indicating that the data may indeed come from a normal distribution. Hence, we can assume that the PM levels follow a normal distribution.
3. If the data didn't look like normal, we could still construct confidence intervals: Since the sample size is 59, much larger than 40, the sample mean, $\bar{X}$, is approximately normal due to the Central Limit Theorem.

Class exercise

Green environment supporters claim that the average daily PM10 level exceeds the upper limit of 50mg. Evaluate this claim using appropriate confidence intervals or bounds using the data above.

	Confidence bounds		Confidence Intervals	
α	LB	UB	LL	UL
0.01	50.43851	52.74149	50.31502	52.86498
0.05	50.77583	52.40417	50.61986	52.56014
0.1	50.95566	52.22434	50.77583	52.40417

A Large-Sample CI for a Population Proportion

X : # of successes in a random sample with $n \ll$ population size

p : proportion of 'success' in a population

$X \sim \text{bin}(n, p)$

For large n with $np \geq 5$ and $n(1-p) \geq 5$, X is approximately normal.

$$E(X) = np$$

$$V(X) = np(1 - p)$$

A Large-Sample CI for a Population Proportion

$$X \sim \text{bin}(n, p) \quad \begin{aligned} E(X) &= np \\ V(X) &= np(1 - p) \end{aligned}$$

For large n with $np \geq 5$ and $n(1-p) \geq 5$, X is approximately normal.

Given this what is the distribution of $\hat{p} = \frac{X}{n}$?

Parameters?

Now find the corresponding confidence interval for p using this information.

Estimate the variance of X if necessary.

Estimate standard deviation of p

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} \quad \rightarrow \quad \hat{\sigma}_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

An approximate confidence interval for a population proportion p with confidence level approximately $100(1-\alpha)\%$ is:

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

which is a traditional CI used for decades. This always needs:

$$n\hat{p} \geq 10 \quad \text{and} \quad n(1-\hat{p}) \geq 10$$

Example

100 architecture students in different universities in turkey are sampled, and it is found that 60 of them are female and 40 are male. Construct an approximate 95% CI for p , where p is the probability of having a female architecture student.

$$0.6 \pm 1.96 \sqrt{\frac{0.6 \times 0.4}{100}} = (0.5040, 0.6960)$$

Estimate standard deviation of p

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} \rightarrow \hat{\sigma}_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

An approximate confidence interval for a population proportion p with confidence level approximately $100(1-\alpha)\%$ is:

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

which is a traditional CI used for decades. This always needs:

$$n\hat{p} \geq 10 \text{ and } n(1-\hat{p}) \geq 10$$

However, the intervals may cover the true proportion p in less than $100(1-\alpha)\%$ of the intervals.

New research:

Recent research shows that a slight modification of n and the following estimate of p provide a good confidence interval:

Define $\tilde{n} = n + 4$ and $\tilde{p} = \frac{X+2}{\tilde{n}}$.  Add 2 observations on both outcomes, which increases the uncertainty, so increases the width of the CI.

An approximate confidence interval for a population proportion p with confidence level approximately $100(1-\alpha)\%$ is:

$$\tilde{p} \pm z_{\alpha/2} \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{\tilde{n}}}.$$

$$\tilde{n} = n + 4 = 104$$

Example

$$\tilde{p} = \frac{X+2}{\tilde{n}} = \frac{60+2}{100+4} = 0.596$$

100 architecture students in different universities in turkey are sampled, and it is found that 60 of them are female and 40 are male. Construct an approximate 95% CI for p, where p is the probability of having a female architecture student.

Width of the CIs:

$$\text{Traditional CI: } 0.6 \pm 1.96 \sqrt{\frac{0.6 \times 0.4}{100}} = (0.5040, 0.6960) \quad 0.192039996$$

$$\text{New CI: } 0.596 \pm 1.96 \sqrt{\frac{0.59 \times 0.404}{10}} = (0.4998, 0.6962) \quad 0.196353405$$